

LIME Explanations

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Introduction

LIME (Ribeiro, Singh, Guestrin 2016) explains a single prediction by fitting a sparse linear model locally around the point of interest. Samples are perturbed near the observation, labelled with the black-box model, and weighted by distance; the linear surrogate's coefficients describe the local behaviour of the model.

Prerequisites

Regression / classification; weighted linear regression.

Theory

For observation x : 1. Sample x'_k by perturbing x in feature space. 2. Predict with the black-box: $\hat{y}_k = f(x'_k)$. 3. Weight samples by proximity: $w_k = \exp(-d(x, x'_k)^2 / \sigma^2)$. 4. Fit a sparse linear model g to $(x'_k, \hat{y}_k)$ with weights w_k . 5. Coefficients of g are the local explanation.

Assumptions

A local linear approximation captures the model's behaviour in a neighbourhood; perturbation scheme is appropriate for the feature type (Gaussian for continuous, resampling for categorical, word removal for text).

R Implementation

```
library(lime); library(ranger)

set.seed(2026)
n <- 500
df <- data.frame(
  x1 = rnorm(n), x2 = rnorm(n), x3 = rnorm(n),
  y = factor(sample(c("a", "b"), n, replace = TRUE))
)
df$y <- factor(ifelse(plogis(0.7 * df$x1 - 0.4 * df$x2) > runif(n), "a", "b"))

rf <- ranger(y ~ ., data = df, num.trees = 200, probability = TRUE,
  classification = TRUE)

# LIME setup
explainer <- lime(df[1:450, -4], rf,
  bin_continuous = TRUE, n_bins = 4)
expl <- explain(df[451:452, -4],
  explainer, labels = "a",
  n_features = 3, n_permutations = 1000)

expl[, c("feature", "feature_weight", "feature_desc")]
```

Output & Results

Per-observation feature weights: how strongly each feature pushes the prediction toward the explained class, with a human-readable feature description.

Interpretation

“LIME attributed observation 1’s predicted class ‘a’ to $x_1 > 0.8$ (weight 0.42) and $x_2 < -0.3$ (weight 0.18), providing an interpretable local rule.”

Practical Tips

- LIME is unstable: different runs can produce different explanations due to random perturbations; set a seed and run several times.
- LIME’s linear surrogate is a local approximation; for globally highly non-linear models, explanations drift across neighbourhoods.
- Compare LIME with SHAP on the same observations; large disagreements indicate explanation fragility.
- For text, LIME removes words and recomputes predictions; for tabular, use binned perturbation.
- Weighted sampling bandwidth σ strongly affects results; tune by inspection.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren’t.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Cross-validation, nested CV, bootstrap .632+
- Regularisation: ridge, lasso, elastic net
- Trees, random forests, gradient boosting
- Interpretability and SHAP
- Tabular neural networks with torch
- Imaging and sequence models intro
- tidymodels pipelines
- FDR, knockoffs, replication crisis